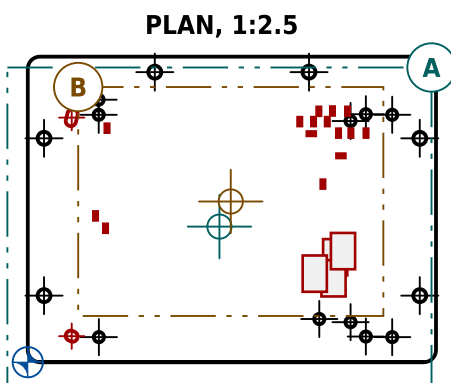
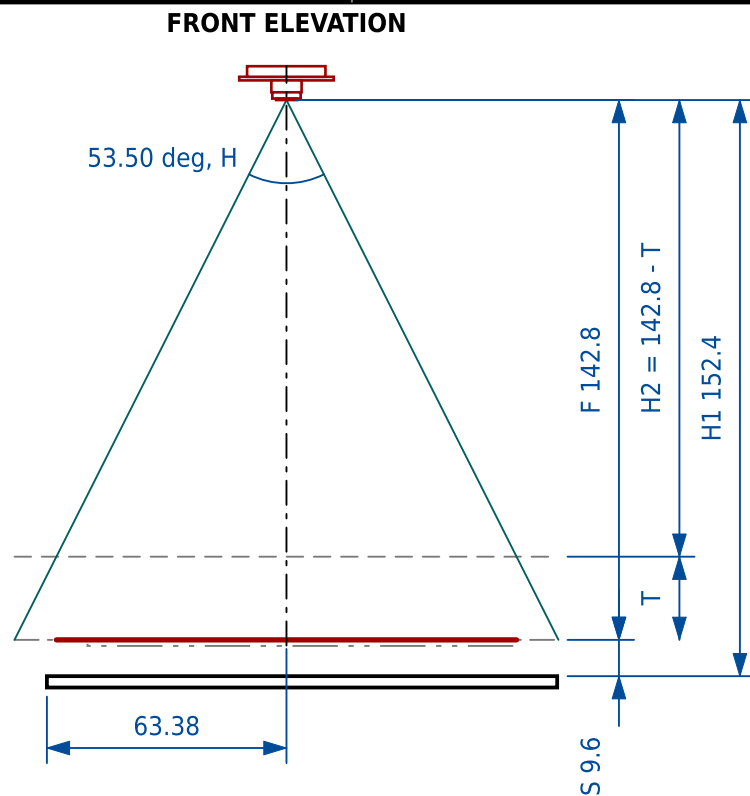
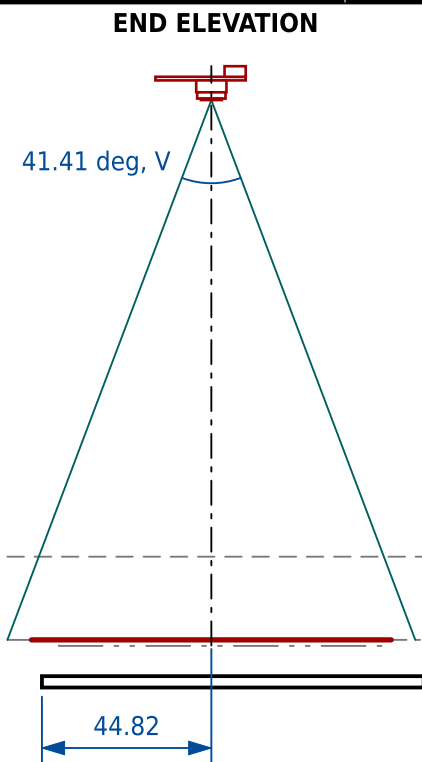




NOTES

1. First angle; the end elevation is seen from the right. One camera: a Raspberry Pi Camera Module v1.3, stock 65 deg lens, looking straight down, reading the whole sensor.
2. Heights are to the lens FACE. The entrance pupil the optics are worked to is ASSUMED behind it by at most the lens's 5.20 mm, so the picture is up to 5.2% larger, never smaller, and the boards no nearer than the close limit.

LENS FACE HEIGHTS, mm: BOARDS IN VIEW AND IN FOCUS

	THE LENS FACE ABOVE	FRAME A	FRAME B	TERM
H1	the mounting plate's face, the standoffs' base	152.4	113.3	S 9.6 +/-0.04
H2	the highest point on the board, as clearance	142.8 - T	103.7 - T	T: MEASURE
F	the demo board's top face, any revision	142.8	103.7	DERIVED

FRAMES, AND WHERE THE LENS AXIS GOES, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y
A	Every board, any revision	140.27 x 105.20	X	63.38	44.82
B	Every LED and 7-seg	100.91 x 75.69	X	67.13	53.12

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- Frame A's target, on the plane F is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- The mounting plate's face, S below the board's top face
- - - The highest point on the board, T above the board's top face: NOT TO SCALE, to measure
- Dimension, extension and leader

NOTES (continued)

11. T: no board file gives a part a height, and nobody publishes one for every revision. Measure it on the boards in use, from the board's top face to the top of its tallest part.

SOURCES

1. Plate geometry: tinytapeout/mounting_plate/plate.py - Outline, placements and standoff; see TT-MP-PLATE.
2. Board and indicator positions: tinytapeout/boards.py - Outlines, connectors, Pmod bodies, LEDs and 7-segment displays, in plate coordinates, and each board's thickness.
3. raspi.tv, 25 May 2013: <https://web.archive.org/web/20260212013652/https://raspi.tv/2013/adapt-your-raspberry-pi-camera-for-close-up-use>
4. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE TT MOUNTING PLATE

Camera Module v1.3, stock 65 degree lens: lens face heights

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-536-g6b746d3	mm	1:2, PLAN 1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-PLATE	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
TT Mounting Plate	plate +/-0.20, LEDs +/-0.10, heights DERIVED		